A System architecture

Semantics belong to plugins; composition and routing to the application; one execution's lifetime to Core

Rho-plugins

capability providers

R analysis · Agent bridge
tables & plots · editor & files
own semantics, native state, provenance and results

Rho application

organizer

select & assemble · layout
shared interaction routing and presentation

Rho-core

optional managed execution

request identity · retained execution facts · resident Host
not on every path

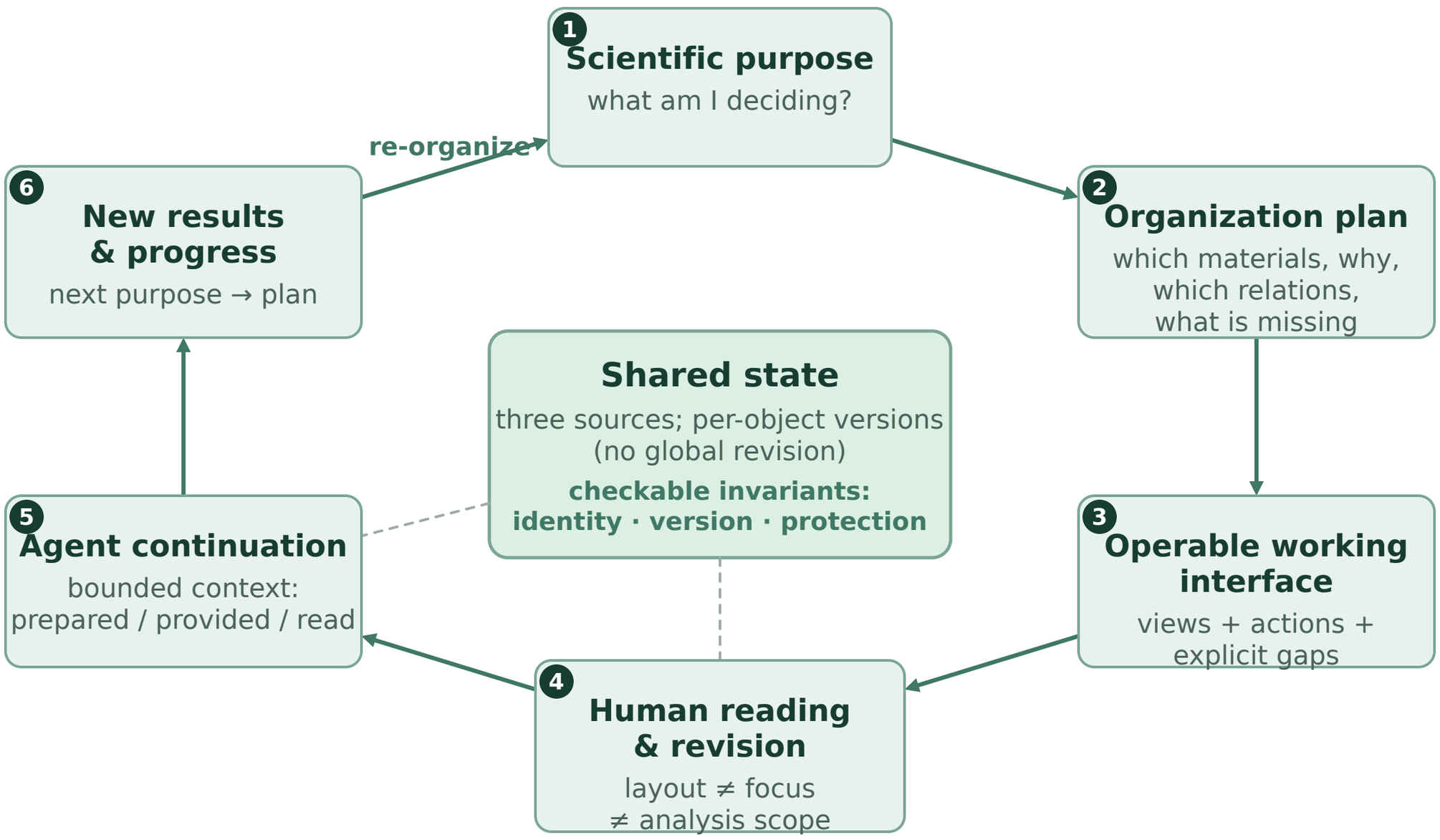
Shared state

actions · resources · selections · views — each with owner, version and provenance
observed computation fact | agent interpretation | human judgment
local service on the machine · no Rho server · the browser has no shell, filesystem or Core access

two entrances read and write the same actions and resources

Researcher

Agent

B The analysis round-trip

The interface and the Agent's context are two presentations of one shared state; versions are per object, not global